

DSA 4060: Recommender Systems
Assignment 1: Semester Project Proposal

NearBite

A Delivery-Aware Restaurant Recommender that Balances Favourite Reorders
with Relevant New Discoveries

Student: Justice Chawanda

Student ID: 670444

Project type: Individual project

GitHub repository: <https://github.com/jpchawanda1/nearbite-recommender>

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A. Project Title and Problem Statement

Title. NearBite: a delivery-aware restaurant recommender for food-delivery customers.

Statement. I will develop a restaurant recommender for a food-delivery app (similar to Glovo in Nairobi or foodpanda in Singapore) that ranks ten restaurants for each customer, using their order history and the behaviour of similar customers. Every suggestion must be able to deliver to the customer's current location, and the list will mix familiar favourites with relevant restaurants they have not tried yet.

Problem and who experiences it. Food-delivery customers must choose from thousands of restaurants on a small phone screen. In the Singapore data used here there are 7,411 restaurants across 78 cuisines. The home screen usually shows city-wide promotions or "popular near you" lists that ignore personal taste, so customers scroll for a long time, reorder the same few places out of habit, or leave the app. Small restaurants are rarely discovered.

Why selection is difficult.

- Most customers have very short histories: 41.5% placed only one order and the median is two, so their tastes are hard to infer.
- Customer-restaurant interactions are extremely sparse (0.037% of the matrix is filled).
- Suggestions are only useful if the restaurant can deliver to the customer's address, so relevance depends on location as well as taste.
- Behaviour is two-sided: 34% of customers' later orders go back to a restaurant they have already used, while the rest are explorations. A good list must support both.

How recommendations help. A ranked, personalised and deliverable shortlist reduces search time, surfaces relevant new restaurants (including smaller ones), and gives each choice a short reason ("because you often order Japanese"), which builds trust.

B. Intended Users and Recommendation Task

Element	Definition in NearBite
Users	Customers of a food-delivery app; the demo uses the Singapore customers in the dataset.
Items	Restaurants (vendors). Menu products are used as item descriptions, not as recommended items.
Inputs	Past orders (restaurant, day, time); current delivery location (geohash cell); optional cuisine likes/dislikes entered by the user; restaurant cuisine, chain, menu and price level.
Output	Top-10 ranked restaurants, split into "Order again" and "Discover" slots, each with a one-line explanation.
Moment	When the customer opens the app or changes the delivery address (on request, real time).
Constraints	Restaurant must be within the delivery radius (≤ 7 km between location-cell centres, which covers 98.7% of historical orders); no duplicates; user-excluded cuisines or restaurants never shown.

User scenario. Amina, a customer in eastern Singapore, opens NearBite at 7 pm on a Friday. She has previously ordered chicken rice twice and bubble tea once, and has marked "no fast food". The system removes restaurants that cannot deliver to her area and fast-food outlets. It then returns ten restaurants: her two chicken-rice favourites under "Order again", and eight "Discover" restaurants ranked by what similar customers order, e.g. a Hainanese stall with the note "Customers who order from your favourite chicken-rice stall also order here". She can remove a suggestion or slide towards "more new places" and the list updates immediately.

C. Dataset Selection and Feasibility

Dataset. Delivery Hero Recommendation Dataset (DHRD), Singapore subset (Assylbekov et al., RecSys 2023). Source: <https://github.com/deliveryhero/dh-reco-dataset> (MIT licence). It contains real foodpanda orders from three months, with pseudonymised identifiers. I downloaded and inspected the Singapore training files (notebook 01_data_inspection.ipynb in the repository).

Quantity	Value	Quantity	Value
Orders	1,709,414	Order lines (products)	3,431,870
Customers	476,150	Restaurants (catalogue / ordered)	7,411 / 7,203
Menu products	1,066,840	Cuisines / chains	78 / 1,854

Period	76 days	Customer-restaurant pairs	1,285,976 (0.037% density)
Orders per customer	mean 3.6, median 2	Customers with ≥ 5 orders	22.1% (105k)

Files and important columns.

- **orders:** customer_id (hashed), geohash (customer location cell), order_id, vendor_id, product_id, day_of_week, order_time, order_day (day index 0 to 75).
- **vendors:** vendor_id, chain_id, geohash (restaurant location), primary_cuisine; latitude/longitude of the cell centre.
- **products:** vendor_id, product_id, name (menu item text), unit_price (scaled).

Interactions and attributes. Interactions are implicit: every order is a customer-restaurant purchase event (repeat orders give a count). There are no explicit ratings. Item attributes: cuisine, chain, location, menu item names (text for TF-IDF) and a relative price level. The three files come from the same source and share vendor_id/product_id keys; I verified that 100% of ordered vendors and products match, so no cross-dataset matching is needed.

Known limitations.

- No ratings, delivery times, opening hours or rider data. Rating-prediction metrics (RMSE/MAE) are therefore not applicable, and the rider extension must use simulated riders.
- Locations are coarse 5-character geohash cells (~4.9 km), so distance is approximate (68% of orders come from the customer's own cell).
- Prices are rescaled and 27.6% of menu items have price 0; 1,499 vendors lack a chain_id and 40 products lack a name. These are handled by filling or ignoring them.
- Restaurant popularity is unusually flat: the busiest restaurant has only 645 orders and the top 10% of restaurants account for just 24% of orders, which suggests the data was sampled. City-wide popularity is therefore a weak signal.
- The official Google Drive folder now asks for a sign-in, so I use publicly shared copies of the same Singapore files that add decoded cell coordinates. The repository documents both routes.

Access and usage. MIT licence permits use and redistribution with attribution. Even so, the ~420 MB of data is not committed: the repository provides scripts/download_data.py and a small sample only. The data contains no names, addresses or contact details. A dataset sample is in Appendix A.

D. Proposed Recommendation Approaches

Every approach produces a score per restaurant. Restaurants outside the delivery radius are removed before ranking, so all methods satisfy the same constraint and are compared fairly.

Baseline: Local popularity (implemented). Suits the problem because ordering is highly local. For each location cell it ranks deliverable restaurants by how often customers in that cell ordered from them, with city-wide popularity as a tie-breaker. Data: orders and geohash only. It gives every new customer a sensible list and sets the bar the other methods must beat. Limitation: identical lists for everyone in an area, and a bias towards already-busy restaurants.

Approach 1: Content-based filtering. Builds a taste profile from the cuisines, chains, price level and menu words (TF-IDF over product names) of restaurants a customer ordered from, then ranks deliverable restaurants by cosine similarity. Data: vendor and product attributes plus each customer's history. It explains recommendations ("you like Japanese") and handles new restaurants with no orders. Limitation: it over-specialises, recommending more of the same cuisine, and cuisine labels are coarse.

Approach 2: Item-based collaborative filtering. Computes restaurant-restaurant cosine similarity from the implicit customer $\times$ restaurant matrix (log-scaled order counts) and scores a restaurant by its similarity to restaurants the customer used. Data: orders only. It captures patterns that labels miss ("people who order from A also order from B") and yields natural explanations. Limitation: weak for customers with one order and for new restaurants, and the matrix is very sparse.

Approach 3: Matrix factorisation (ALS for implicit feedback) and hybrid. Weighted ALS (Hu, Koren & Volinsky, 2008) learns latent taste factors from order counts; the hybrid blends MF, content scores and a

repeat/recency signal, with weights tuned on the validation week. It meets the course requirement for MF or a hybrid and targets both repeat and exploration behaviour. Data: orders plus attributes. Limitation: latent factors are hard to explain, and tuning is computationally heavier.

Extension (weeks 11 and 12): rider ranking. For a ready order, riders are ranked by a transparent weighted score combining closeness to the restaurant, rating, current load and acceptance rate. DHRD has no rider data, so riders are simulated around restaurant locations. This module is presented as an illustration and is not one of the graded recommendation approaches.

Preliminary evidence (test week, 104,890 customers, K = 10).

Method	Precision@10	Recall@10	NDCG@10	Coverage
Global popularity, no delivery filter	0.0001	0.0004	0.0002	0.001
Local popularity (baseline)	0.0063	0.0508	0.0233	0.044
Personal repeat frequency (reference)	0.0416	0.3059	0.2437	0.934

Location alone raises NDCG more than tenfold over a city-wide list. None of the unfiltered city-wide lists could be fully delivered to any customer, which confirms the delivery constraint is necessary. Simply repeating past restaurants is strong, but it finds only 3.5% of the new restaurants customers tried, which is the gap the CF, MF and hybrid approaches must close.

E. Evaluation Plan

Data separation. A temporal split mimics real use: models learn from the past and predict the following week. Development uses days 0 to 61 for training and days 62 to 68 for validation (all tuning). The final test retrains on days 0 to 68 and evaluates once on days 69 to 75. For each customer active in the target week, the relevant set is the restaurants they ordered from that week, and the list is generated for the location of their first order that week. Warm customers (81.6%) and new customers (18.4%) are reported separately.

Measure	What it assesses and why the data supports it
Precision@10, Recall@10, Hit-rate@10	Whether restaurants the customer actually ordered from next week appear in the top 10. Held-out orders provide the ground truth.
NDCG@10	Ranking quality: relevant restaurants near the top score more, which matters because only the first few fit on screen.
Exploration recall@10	Recall restricted to restaurants the customer had never used, i.e. real discovery. Computable because full order histories are available.
Catalogue coverage	Share of the 7,411 restaurants that ever get recommended, which measures popularity bias and fairness to small restaurants.
Cuisine diversity	Distinct cuisines per list, which guards against repetitive lists (uses <code>primary_cuisine</code>).
Feasibility rate, response time	Share of lists that are fully deliverable (must be 100%) and ms per request (target < 200 ms for the interface).

Comparison. RMSE/MAE are not used because the data has no ratings. Every method is scored on the same customers, and the per-customer NDCG differences against the baseline are tested with a paired bootstrap (95% CI). The evaluation harness is already implemented and reproducible (scripts/run_baseline.py).

F. Challenges and Responsible Design

Challenge	Initial response
New customers (18% of test customers)	Local popularity for the first visit, plus an optional quick cuisine picker that seeds the content-based profile.
New restaurants with no orders	Content-based scores from cuisine and menu text; a small "new nearby" slot.
Sparse interactions (0.037%)	Model at restaurant level (not dish level), use implicit-feedback ALS with log-scaled confidence, and filter by minimum support in item-CF.
Popularity bias and repetitive lists	Monitor coverage and diversity, reserve "Discover" slots, and cap the number of restaurants per cuisine or chain in a list.
Changing preferences	Weight recent orders more (time decay) and retrain weekly; time of day (lunch vs dinner peaks) is a candidate feature.
Unavailable or ineligible restaurants	Hard delivery-radius filter (100% feasible lists). Opening hours are not in the data; this is noted as a limitation.
Privacy	Pseudonymous IDs and coarse location only; no raw data in the repository; the interface shows a customer only their own history.

Understanding and influencing recommendations. Every recommendation carries a short reason ("similar to X you ordered", "popular in your area", "matches your liking for Indian"). Users can hide a restaurant or cuisine, set cuisine preferences, choose a maximum distance, and move a "favourites vs new places" slider that changes the balance between repeat and discovery scores.

G. Implementation Plan and Responsibilities

Tools. Python 3, Pandas, NumPy, SciPy (sparse matrices), scikit-learn (TF-IDF, cosine similarity), implicit (ALS), Matplotlib, Jupyter Notebook, pytest, Streamlit (interface), Git and GitHub.

Responsibility. This is an individual project, so I am responsible for every task and for understanding the complete system end to end.

Week	Main tasks	Expected output	Status
1-2	Choose problem, inspect data, repository, evaluation harness, baseline	Proposal, README, inspection notebook, baseline results	Done
3-4	Data preparation, feature building, content-based prototype	Content-based model + first comparison	Planned
5-6	Item-based CF, preliminary evaluation on validation week	CF model, comparison table	Planned
7	Mid-semester examination	None	N/A
8-10	ALS matrix factorisation, hybrid, tuning	MF and hybrid models, tuned weights	Planned
11-12	Streamlit interface, explanations and controls, rider extension, final test evaluation, documentation	Working app, final report	Planned
13	Presentation and demonstration	Slides, live demo	Planned

References

- Assylbekov, Y., Bali, R., Bovard, L., & Klaue, C. (2023). Delivery Hero Recommendation Dataset: A novel dataset for benchmarking recommendation algorithms. Proceedings of the 17th ACM Conference on Recommender Systems (RecSys '23), 1042-1044. <https://doi.org/10.1145/3604915.3610242>
- Delivery Hero. (2023). dh-reco-dataset [Data set, MIT licence]. GitHub. <https://github.com/deliveryhero/dh-reco-dataset>
- Hu, Y., Koren, Y., & Volinsky, C. (2008). Collaborative filtering for implicit feedback datasets. Proceedings of the 8th IEEE International Conference on Data Mining, 263-272.
- Järvelin, K., & Kekäläinen, J. (2002). Cumulated gain-based evaluation of IR techniques. ACM Transactions on Information Systems, 20(4), 422-446.
- Sarwar, B., Karypis, G., Konstan, J., & Riedl, J. (2001). Item-based collaborative filtering recommendation algorithms. Proceedings of the 10th International World Wide Web Conference, 285-295.

Appendix A. Dataset Sample

Table A1. Eight random order lines (orders joined with vendor cuisine and product name/price).

order_id	customer	geohash	vendor	cuisine	product name	price	dow	time	day
1188376	54dc6d3cbf	w21z6	2ee5659d	chinese	1022 Marmite Chicken Bento Set 妈蜜鸡丁套餐	0.0048	6	18:13:26	35
673460	f68655bae9	w21zb	f49aea1a	japanese	3.Salmon + Pork Katsu Bento 三 文鱼+炸猪排饭盒	0.0028	5	16:27:36	62
160126	3df0afd086	w21zu	49de578b	asian	Vegetarian Bee Hoon	0.0048	3	08:48:05	4
1743166	969abe0e1d	w21zu	fbb6785	burgers	FISH'N CRISP CROISSAN'WICH® Meal	0.0028	1	09:29:28	23
1913308	2bdbc42d2c	w23bh	fc64fe17	chicken rice	Roasted Chicken Rice with Beverage 烧鸡饭和水	0.0028	6	16:44:32	49
1089155	7f089e7e3f	w21ze	ddcf22f1	chicken rice	Signature White Chicken for 1 Pax	0.0024	3	12:31:58	74
1387829	c179bdbc9e	w21zt	5aa428a9	chinese	Fried Chinese Cabbage & Fungus with Garlic 木耳大白菜	0.0032	5	13:07:46	55
242402	d7e2721d7b	w21zg	85443216	indian	B02. Briyani Chicken	0.0044	0	11:43:43	64

Table A2. First five rows of vendors.

vendor_id	chain_id	geohash	primary_cuisine	latitude	longitude
b160c319	d2786168	w21z6	mexican	1.2964	103.8208
9c8f010e	d2786168	w21ze	mexican	1.3403	103.8647
03eb25e1	5055ab25	w21ze	bak kut teh	1.3403	103.8647
3613129a	8984acb6	w23b1	italian	1.4282	103.7769
0946c9e5	1e3a2913	w21z4	bak kut teh	1.2524	103.8208

Table A3. First five rows of products.

vendor_id	product_id	name	unit_price
15bbf316	59099e089514	Meat & Seafood Combo 海鲜套餐	0.0036
e7b24dc0	52e1017fdbd9	Sausage Egg McMuffin® Meal	0.0024
e7b24dc0	fc67110cd932	Scrambled Egg Burger w Chicken Meal	0.0024
7112a20b	d3e7708c2bc9	Maguro Sushi	0.0012
6137ef21	bf0d5ac0e03c	Family Meal - Prosp Chic TwLEVMx2 + ChBurgerHM	0.01

Figure A1. Customer activity (left) and the flat restaurant popularity curve (right).

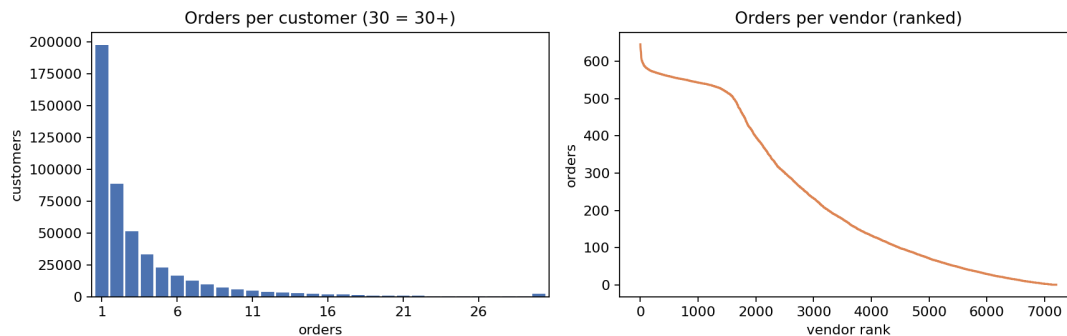


Figure A2. Top cuisines and lunch/dinner ordering peaks.

